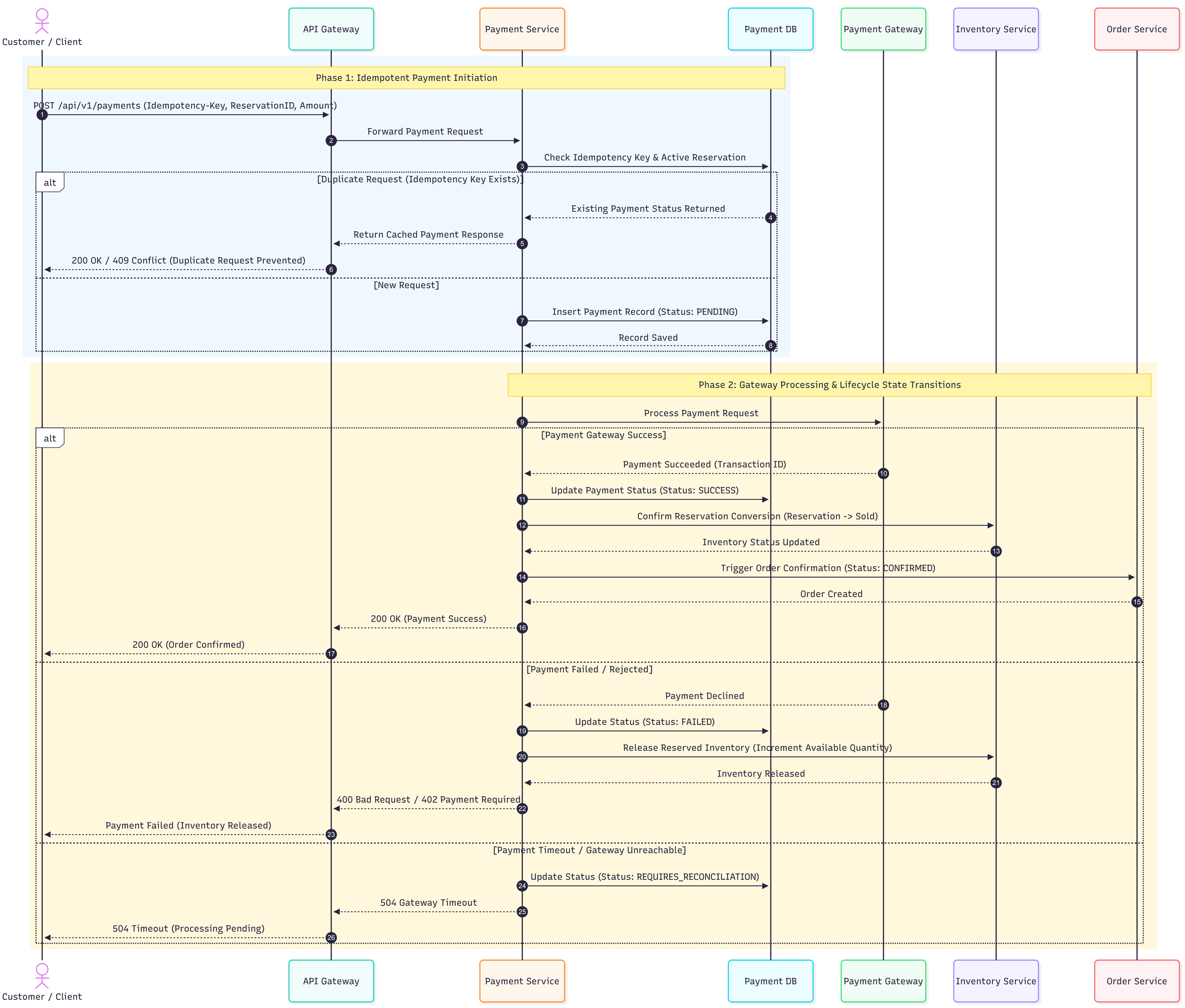

Customer / Client
API Gateway
Payment Service
Payment DB
Payment Gateway
Inventory Service
Order Service
Phase 1: Idempotent Payment Initiation
POST /api/v1/payments (Idempotency-Key, ReservationID, Amount)
1
Forward Payment Request
2
Check Idempotency Key & Active Reservation
3
alt
[Duplicate Request (Idempotency Key Exists)]
Existing Payment Status Returned
4
Return Cached Payment Response
5
200 OK / 409 Conflict (Duplicate Request Prevented)
6
[New Request]
Insert Payment Record (Status: PENDING)
7
Record Saved
8
Phase 2: Gateway Processing & Lifecycle State Transitions
Process Payment Request
9
alt
[Payment Gateway Success]
Payment Succeeded (Transaction ID)
10
Update Payment Status (Status: SUCCESS)
11
Confirm Reservation Conversion (Reservation -> Sold)
12
Inventory Status Updated
13
Trigger Order Confirmation (Status: CONFIRMED)
14
Order Created
15
200 OK (Payment Success)
16
200 OK (Order Confirmed)
17
[Payment Failed / Rejected]
Payment Declined
18
Update Status (Status: FAILED)
19
Release Reserved Inventory (Increment Available Quantity)
20
Inventory Released
21
400 Bad Request / 402 Payment Required
22
Payment Failed (Inventory Released)
23
[Payment Timeout / Gateway Unreachable]
Update Status (Status: REQUIRES_RECONCILIATION)
24
504 Gateway Timeout
25
504 Timeout (Processing Pending)
26
Customer / Client
API Gateway
Payment Service
Payment DB
Payment Gateway
Inventory Service
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